

TableTidier

Extracting and Harmonising Tabular Data for Secondary Research

Jesus A. Rodriguez Perez @ Datasky.uk

David McAllister @ University of Glasgow

The Problem - Manual Data Extraction

By **data extraction**, we refer to the exercise of linking each data point in a table to its headers. Then **each data point** has enough information to be **interpreted within its** intended **context**.

Most data in research documentation is **contained within tables**. Thus, extracting data from them is a recurring exercise, which is often done manually, which is **expensive, time consuming** and **error prone**.

Programmatically extracting data from tables is **non-trivial**. Complex table structures emerge from constrained space and reliance on subtle **visual cues** tailored for human interpretation.

Opportunities / Vision

A programmatic platform dedicated to extraction and management of data with extra possibilities:

1. **Ease and support reliable data extraction from documents, reducing human intervention.**
2. **Curated research repositories**, fostering reuse of extracted data
3. Enhanced functions **augmented by ontologies** (UMLS), such as search, recommendations, etc.
4. Providing **APIs** to applications that interrogate and report using the available data, and its derivatives.
5. Sensemaking and **data visualisation** tools to describe our data collections.

Introducing TableTidier

Initially born (many years ago) **as a project with strong R dependencies** to cleanup data, and prepare it for statistical analysis.

Evolved organically over time with support for extra functionality which streamlined the data extraction processes of our **Public Health group @ Glasgow University**

As the project grew organically more complex, further development and maintainability concerns, motivated us towards a simpler design using **popular web application frameworks** (Node.js/React, and now NextJS).

TableTidier Core Functionality

Web app providing an end-to-end 4-step workflow for data extraction from **.html** tables:

1. **Uploading & Viewing:** Import .html documents to extract tables. Then organise them into collections.
2. **Annotating & Editing:** Annotate the structure of a table via an intuitive interface to extract the data. Fix any errors with a WYSIWYG editor.
3. **Metadata:** Assign UMLS concepts to headings with an AI-assisted system enriching strings data with CUIs (Concept Unique Identifiers).
4. **Exporting:** Download your curated data as a clean, structured JSON or CSV file, ready for analysis or integration into other systems.

1. Uploading & Viewing

Collections

- Tables are uploaded into collections.
- Collections group together tables as organised by the user. The purpose and topic of the collection can be specified in it's name and description.

Tables are extracted from .html files:

- On uploading, TableTidier automatically detects the presence of <html> tags (With extra logic to deal with other structural parameters such as rowspan; colspan)
- Tables are transformed into 2-dimensional Arrays, that are easily managed by the application.

2. Annotating & Editing

In order to extract the data, the user specifies the cells that are being used as headings, and how these **headings** are **grouped together** into **concept groups**.

Concept groups represent **categories and subcategories** of headings. TableTidier, allows also for these dependencies to be explicitly specified via a fast and intuitive drag and drop system.

The result of this exercise is a downloadable file, where every data point is appropriately connected to its headings **in a matter of seconds**.

3. Metadata

In order to augment our data and enable future functionality, we employ a metadata linkage system*, that connects each concept in the table to Concept Unique Ids in the UMLS ontology.

This will allow more advanced functionality such as:

- Fuzzy searches, where concepts may be matched semantically as closely related within the ontology, across different studies; tables.
- Collection summaries, reporting how specific aspects of the selected studies/tables cover certain topics of interest.

*built using a medical LLM and a vectorial database (Qdrant)

Demo



TableTider 11419425_1.html

Back to Collection Collection: 4f67164-1937-4292-840e-172169950279 Created: April 21, 2025 at 12:21 AM

Annotation Dashboard Extracted Data Metadata

Export Table HTML

CHARACTERISTIC	TRIOFIBAN (N=2398)	ABICXIMAB (N=2410)
Age (yr)	62.6411	62.6411
Male sex (%)	74	73
Weight (kg)	86.1	86.1
Diabetes (%)	23	23
Hypertension (%)	64	65
Smoker (%)	65	64
Medical history (%)		
Coronary-artery bypass grafting	17	17
Percutaneous coronary revascularization	29	30
Myocardial infarction	40	39
Cerebrovascular accident	3	3
Primary target vessel (%)		
Left anterior descending coronary artery	38	34
Right coronary artery	33	34
Left circumflex artery	22	25
Bypass graft	6	5
Retrograde vessel	5	5

Study arms

CHARACTERISTIC

TRIOFIBAN (N=2398)

ABICXIMAB (N=2410)

Characteristic

Medical history (%)

Primary target vessel (%)

CHARACTERISTIC

Age (yr)

Male sex (%)

Weight (kg)

Diabetes (%)

Hypertension (%)

Smoker (%)

Coronary-artery bypass grafting

Percutaneous coronary revascularization

Myocardial infarction

Cerebrovascular accident

Left anterior descending coronary artery

Right coronary artery

Left circumflex artery

Bypass graft

Retrograde vessel

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Annotation Dashboard Extracted Data Metadata

Filter Related Data Export Table HTML

Concept Groups

study arms

CHARACTERISTIC

TRIOFIBAN (N=2398)

ABICXIMAB (N=2410)

CHARACTERISTIC

Medical history (%)

Primary target vessel (%)

CHARACTERISTIC

CHARACTERISTIC

Age (yr)

Male sex (%)

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Right coronary artery

Left circumflex artery

Bypass graft

Retrograde vessel

All is work in progress, as features get integrated into the system.

We have a strong imperative to give the **best UI/UX experience** possible to assist the users, and make it an enjoyable experience.

Future Work - Short Term

- Upload of **PDF** documents and **automatic conversion into HTML**
- Handling of **multiple tables inside an .html** on upload.
- Notion of “**document**” that groups multiple tables, along with the detection of **document metadata** such as ids (e.g. PMIDs)
- **Search** across multiple collections/documents/tables, with **free text**, and **semantic similarity**.
- Creating a **new collection** out of search results, to enable **re-use of existing tables**
- **Visual representations of the collections** which can **highlight the strengths and weaknesses** of a study, by leveraging the semantic concepts and their interconnections.

Future Work - Long Term

- Browser plugin app, with a “**send to collection**” option when right clicking on a table, and/or document within your browser
- Steering into an Systematic Literature Review platform with data extraction; augmentation and AI assistance at its core.

Thanks for listening!

Want to propose / discuss ideas?

Are you interested in using TabletIdier?

We would love to hear from you, so please reach out:

Jesus A. Rodriguez Perez @ Datasky.uk
jesus@datasky.uk